Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Draw Phase Portraits

1. Determining the Isoclines and Nullclines

Consider the *linear homogeneous system of 1st-order ODEs with constant coefficients*: $\begin{aligned} x' &= v \\ v' &= -x - v \end{aligned}$

The solutions x and v are unknown but we can determine the isoclines and nullclines. Then, we can analyze the solution behaviors by drawing the *vector field*, an x-v plane with direction vectors.

a. An *isocline* is any line or curve along which the direction vectors on a vector field all have the same slope. For some constant m, let $\dfrac{v'}{x'} = m$. Solve for v. Then, write it in slope-intercept form.

b. A *nullcline* is a special case of isocline. For each case of m, determine and explain why whether it describes the x-nullcline or the v-nullcline on an x-v plane. Then, write the nullcline functions.
   i. $m \to 0$
   ii. $m \to \pm\infty$

c. Consider the transformation $v = x'$.
   i. Convert the given linear system of 1st-order ODEs into a 2nd-order ODE.
   ii. Solve for the general solution of the 2nd-order ODE determined in Part (c-i).

d. Explain what happens as $t \to \infty$ of the general solution determined in Part (c-ii) and how this connects to the intersection of the x-nullcline and v-nullcline.

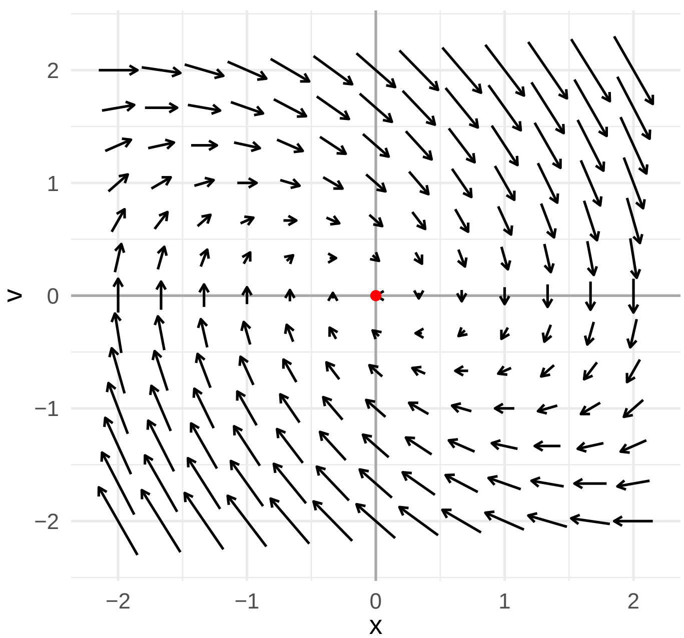
2. Drawing a Vector Field and Phase Portrait

Consider the same linear system from Problem (1): $\begin{aligned} x' &= v \\ v' &= -x - v \end{aligned}$

A *vector field* (also known as *direction field*) is a graphical representation that shows the direction and rate of change of solutions to a system of 1st-order ODEs at every point in the x-v plane (also known as the *phase plane*).

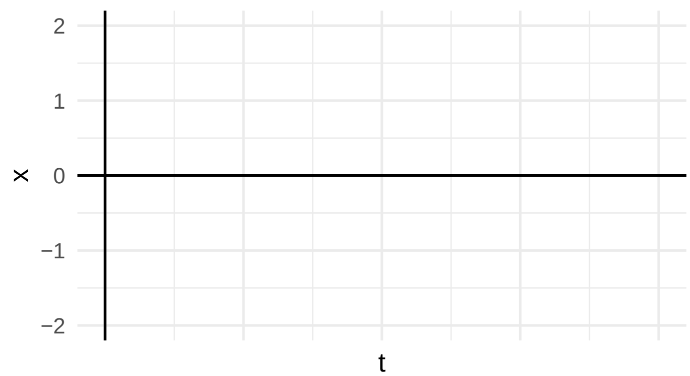
A *phase portrait* is a graphical representation of all possible solution trajectories of a system of 1st-order ODEs in the phase plane.

a. The vector field for the given system of ODEs is shown below. Given the nullclines from Problem (1b), draw these nullclines on the phase plane with the appropriate labels. Then, draw a solution curve starting from the point $(x, v) = (1, 2)$.

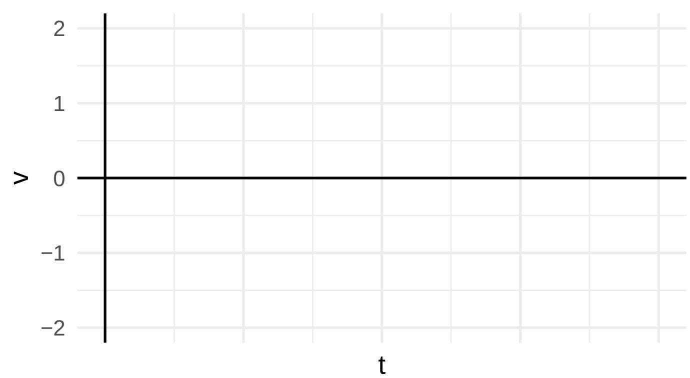


b. Using the general solution determined in Problem (1c), determine the specific solution given the initial conditions $x(0) = 1$ and $v(0) = 2$.

c. Use Parts (a) and (b) to draw an approximate solution curve in the blank t-x plane.



d. Use Parts (a) and (b) to draw an approximate solution curve in the blank t-v plane.



e. Explain why the structure of the specific solution determined in Part (b) is consistent with the solution behaviors shown in Parts (c) and (d).